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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	High Five
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4, 8, 10
Project / Solution Name	CareerLens AI
Name (Leader)	Lai Jie Ying
Email & Phone (Leader)	+60 11-2097 0647, jieying.lai47@gmail.com
University Affiliation(s)	Universiti Tunku Abdul Rahman (UTAR)
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

According to Anthropic, rapid advancements in artificial intelligence (AI) are transforming existing jobs and changing the skills required in the working environment. As AI continues to reshape the labour market, individuals will face increasing uncertainty about which jobs are safe and what skills they need to develop. Traditional career guidance and learning pathways may struggle to keep pace with these changes. This creates difficulty for individuals to make career decisions and identify suitable reskilling opportunities.

These challenges affect working adults, fresh graduates, job seekers and career switchers. Fresh graduates may struggle to identify future-proof career opportunities, while working adults face pressure to continuously adapt their existing skills. Högemann et al. (2025) highlighted the pressure experienced by young professionals in Research and Development (R&D) and Information Technology (IT) to adapt to changing AI-related workplace requirements. Without up-to-date career guidance, individuals may find it difficult to find a suitable job, identify their skill gaps and plan learning pathways.

These challenges also create significant social and economic issues. Uncertainty about changing job requirements may create job insecurity and technostress. Also, Kamrozzaman et al. (2025) highlighted obstacle such as limited digital literacy, financial constraints and inadequate digital infrastructure may widen inequalities in access to reskilling opportunities. This issue may widen existing inequalities in access to AI-related learning opportunities which bring difficulty to the workers less prepared for an AI-driven workforce

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

CareerLens AI is an AI-powered career development and workforce planning platform designed to help job seekers and HR professionals navigate the AI-driven job market. For job seekers, it analyses CVs and skills questionnaires to identify skill gaps, increase job opportunities and recommend personalised reskilling pathways. For HR roles, it supports skills-based candidate screening, identifies suitable talent, and provides strategic workforce forecasts and corporate expansion recommendations. The platform also links career development to relevant UN Sustainable Development Goals (SDGs).

CareerLens AI applies AI in certain function. Job matching uses 16-dimensional skill vectors and cosine similarity, while course recommendations apply TF-IDF and cosine similarity to address skill gaps. Predictive modelling powers the AI Job Disruption Radar to estimate occupational changes. Fine-tuned Qwen2.5 models support interview question generation, response evaluation, Totem Spirit career personality insights, and the Lumi AI chatbot. For employers, the Strategic Workforce & Expansion Advisory Engine combines workforce displacement datasets, DOSM salary benchmarks, CAGR-based simulations, and a Qwen2.5-7B LoRA model to forecast talent demand, automation risks, and salary trends through 2030. It also generates workforce planning and business expansion recommendations aligned with SDG 8.

In RunPod workshop, API integration is learned to connect the AI model with the CareerLens AI Flask backend. Besides, deployment testing also helps to evaluate responsiveness of the request from AI endpoint. Replit workshop enabled responsive interface development, real-time previews, and Git integration.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

1. Frontend Technologies

HTML5, CSS3, JavaScript and Tailwind CSS are used to develop responsive Talent and HR dashboards. Leaflet.js supports interactive job mapping, while HTML5 Canvas and SVG enable visualisations such as the Future Skills Radar.

2. Backend Frameworks

Python 3.11 and Flask 3.x provide RESTful APIs for job matching, candidate screening, skills assessment, course recommendations and workforce analytics. Role-Based Access Control (RBAC), CSRF protection and session management support system security. The pypdf and pdfplumber libraries enable CV extraction and processing.

3. Generative AI and Machine Learning

Google Gemini API supports AI-powered career guidance, contextual assessments and interview question generation. Qwen2.5 models are fine-tuned using Unsloth and LoRA/QLoRA techniques on RunPod GPU infrastructure. TF-IDF and Cosine Similarity enable skills-based job and course recommendations.

4. Databases and Data Sources

SQLite 3 stores user information, applications and access records. Compressed JSON knowledge bases and in-memory indexes support efficient data retrieval. O*NET 29.0, JobStreet Malaysia, DOSM Salaries & Wages Survey 2025 and structured course datasets provide occupational skills, job opportunities, salary benchmarks and learning resources.

5. Cloud Platforms, APIs and Version Control

Replit supports responsive interface development, real-time previews, Git integration and web application deployment. RunPod provides GPU computing resources and JupyterLab for AI model training and fine-tuning. The Remotive Remote Jobs API supplies updated remote job opportunities, while GitHub supports version control and team collaboration.

Refer Appendix 1 for *simple system block diagram*

For official SDC sponsor-provided GenAI API or Cloud Platform

1. Replit: Replit was used to optimise the web application's responsive design across different devices which is to ensure a consistent user experience on desktops, tablets and mobile phones. Its real-time preview enabled developers to immediately visualise and test interface changes without repeatedly pulling code from GitHub. Additionally, Git integration facilitated version control and collaboration among team members.

2. RunPod: RunPod provided a cloud-based GPU computing environment where the team accessed JupyterLab to train and fine-tune AI models. This enabled experimentation with model training workflows and supported the development of specialised AI capabilities for CareerLens AI.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathize: Hays (2025) reported that only 26% of surveyed Malaysian received AI training for work while Chan (2026) found that 33% of surveyed U.S. hiring managers believed ATS overly on keyword matching which highlighted reskilling gaps and recruitment inefficiencies.

Define: AI-driven job uncertainty, skill mismatch, limited personalised reskilling guidance and HR challenges in identifying qualified candidates that support the SDGs.

Ideate: Proposed AI-driven job trend analysis, skills-based career matching, personalised reskilling pathways, SDG mapping, blind candidate screening and workforce analytics.

Prototype: Separate Talent and HR dashboards using Flask, REST APIs and datasets. The project integrates job recommendations, skills verification, learning pathways, candidate screening and workforce planning. Foundation for cloud deployment is also established with integration of Malaysian employment and training platforms

Test: 14 automated test suites passed in validating system functionality, security, and accessibility. Responsive dashboard layouts were also tested to ensure usability across different devices for both job seekers and HR professionals.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target SDG 4.4: support employement-related skill development through personalised reskilling guidance and course recommendations Target SDG 8.5: support job readiness to identify suitable job and assist HR roles in evaluating candidates' qualifications Target SDG 10.2: provide accessible learning course recommendations for individuals from different backgrounds
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to the *AI drives decline in software development roles, but Malaysia holds strong as key global skill hub*. (2025), only 26% of surveyed Malaysian organisations provided AI training or support, while 64% reported moderate to severe skills shortages. Additionally, a U.S. hiring survey reported that 33% of hiring managers believed Applicant Tracking Systems (ATS) overemphasised keyword matching rather than overall candidate suitability (Chan, 2026).

Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Average processing time (seconds) per batch of 50 candidate profiles. 2. ATS false-negative rate, measured as the percentage of qualified candidates incorrectly rejected by keyword-based screening. 3. Average time (minutes) required for job seekers to identify suitable reskilling or upskilling courses that match their identified skill gaps. 4. Multi-label F1-score (%) measuring how accurately CareerLens AI maps job roles, occupational skills and recommended courses to relevant UN Sustainable Development Goals (SDGs).

Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. Achieve at least an 80% reduction in initial candidate screening time compared with manual screening according to *Eye-tracking: Recruiters average 7.4 seconds reviewing a résumé - HR daily advisor*(2018)'s reported average of 7.4 seconds per resume. 2. Fofanah (2026) reported a 47.8% false-negative rate in a controlled simulation where keyword-based screening incorrectly rejected 239 out of 500 qualified candidates. Based on this reference baseline, CareerLens AI aims to achieve a 65% reduction in false-negative rate to 16.73% or lower compared using traditional keyword filtering. 3. Achieve a 75% reduction in the time required for job seekers to identify suitable reskilling courses through personalised skill-gap analysis and targeted course recommendations. *The importance of trust in online education and eLearning* (2019) reported that nearly 70% of consumers spent at least 1–2 hours or more researching e-learning programmes before committing. Using a baseline of 120 minutes, CareerLens AI aims to reduce course discovery time to 30 minutes or less. *4.* Achieve a multi-label F1-score of at least 90% in mapping job roles, occupational skills, and recommended courses to the 17 UN Sustainable Development Goals. The study from Kharlashkin et al. (2024) reported an F1-score of 78.6% using the BART model which act as the reference baseline.

References

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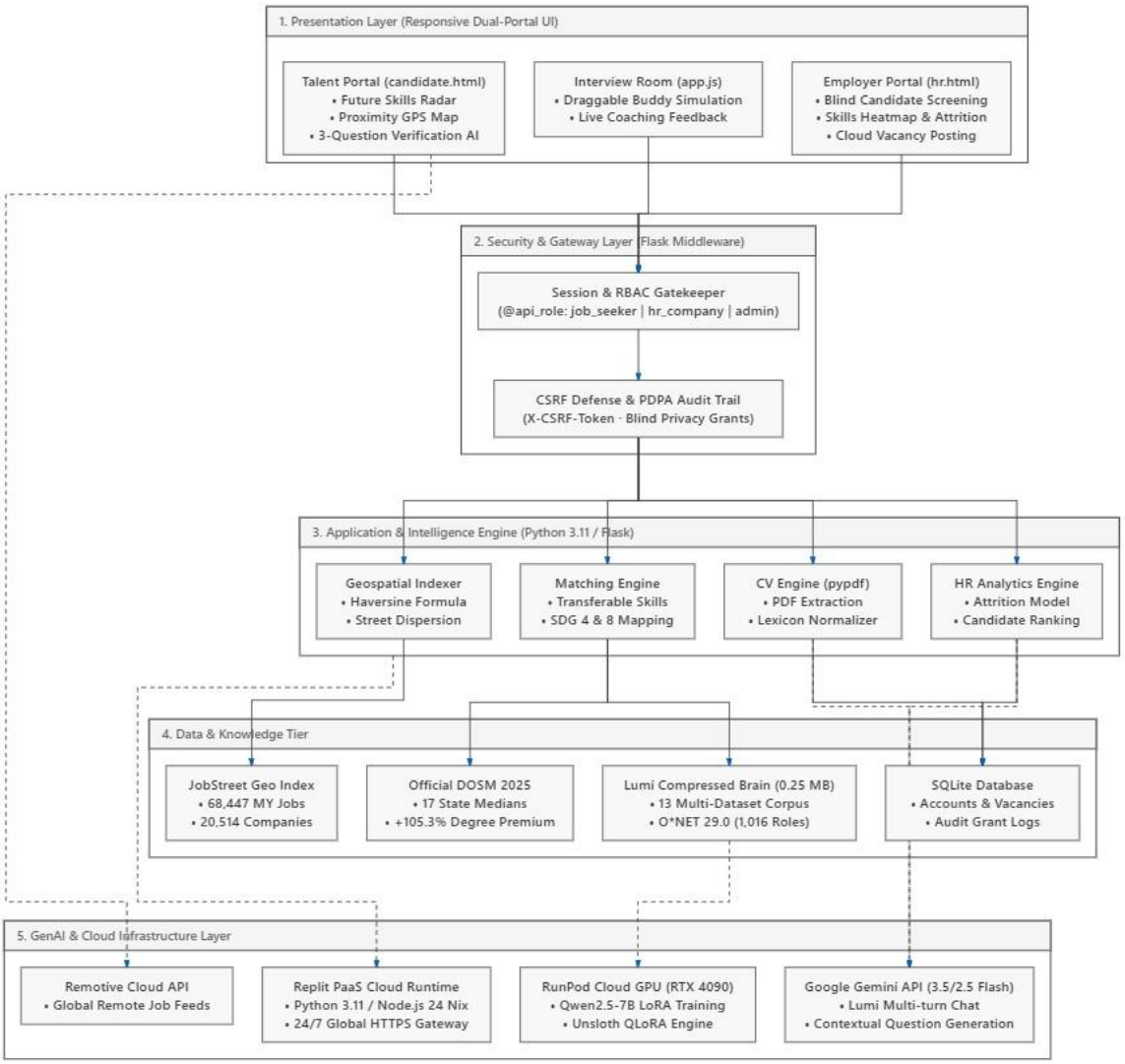
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Appendix 1



system block diagram

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☒ Akk team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.11	Code	PSF License
Flask 3.x	Code	BSD-3-Clause
Tailwind CSS 3.x	Code	MIT
Leaflet.js 1.9.4	Code	BSD-2-Clause
Chart.js 4.x	Code	MIT
pypdf 4.x	Code	BSD-3-Clause
pdfplumber	Code	MIT
fpdf2 2.7.x	Code	LGPL-3.0
Waitress 3.0.2	Code	ZPL-2.1
SQLite 3	Code	Public Domain
openpyxl 3.1.x	Code	MIT
pandas 2.x	Code	BSD-3-Clause
Unsloth	Code	Apache-2.0
scikit-learn (if used)	Code	BSD-3-Clause

O*NET 29.0 / 30.3 Database	Dataset	CC BY 4.0
DOSM Salaries & Wages Survey 2025	Dataset	Malaysia government open-data terms
JobStreet Malaysia All Job Dataset (Kaggle)	Dataset	CC0
AI Impact on Job Market (2024–2030)	Dataset	CC BY-SA 4.0
AI Workforce Displacement Global (2020–2026)	Dataset	ODbL
EdX University Courses Dataset 2021	Dataset	CC0
IBM HR Analytics Employee Attrition & Performance	Dataset	Apache-2.0
Gender Pay Gap Dataset	Dataset	Copyright © Original Authors
Updated Resume Dataset (Kaggle)	Dataset	CC BY-SA 4.0
Qwen2.5-7B-Instruct	Model	Apache-2.0
Qwen2.5-1.5B-Instruct	Model	Apache-2.0
Google Gemini 2.5 Flash (API model)	Model	Google Generative AI terms
Google Gemini 3.5 Flash (team-listed; verify identifier)	Model	Google Generative AI terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Lai Jie Ying]

[25/10/2026]